

Hidden Layers	Neurons	Learning Rate	Accuracy	Epochs to Converge
2	10	0.001	1.00	609
4	10	0.001	1.00	1214
2	10	0.0005	1.00	935
2	20	0.001	1.00	526
2	10	0.01	1.00	157
2	6	0.01	1.00	142

Question 1:

Accuracy stayed at 100% across every configuration.

- More layers hurt: Increasing from 2 to 4 hidden layers (10 neurons, lr=0.001) took epochs from 609 to 1214 for no accuracy gain.
- More neurons helped: 2 layers with neurons increased from 10 to 20 (lr=0.001) dropped epochs from 609 to 526.
- Fewer, well-placed neurons helped worked best: with lr=0.01, going from (10,10) to (6,6) neurons reduced epochs from 157 to 142.

Best configuration: 2 hidden layers, 6 neurons each, lr=0.01. Fastest convergence (142 epochs), same 100% accuracy.

Question 2:

Holding architecture fixed at 2×10 and changing the learning rate resulted in

- Epochs: 935 (lr=0.0005)
- Epochs 609 (lr=0.001)
- Epochs 157 (lr=0.01)

An inverse relationship, higher learning rate converges in fewer epochs. Accuracy stayed 100% at every rate.

Best learning rate: 0.01

Umais Jamal (538263)

Hidden Layers: 2, Neurons: 10, Learning Rate: 0.001

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

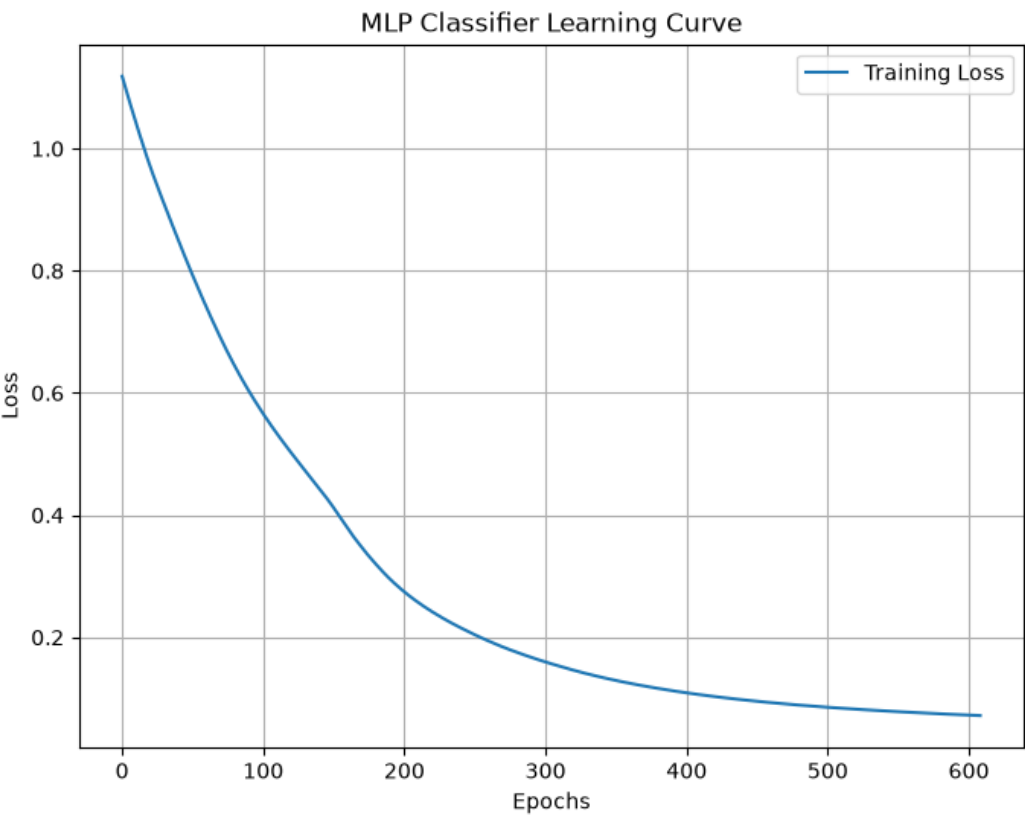
Number of layers: 4

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 609



Umair Jamal (538263)

Hidden Layers: 4, Neurons: 10, Learning Rate: 0.001

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

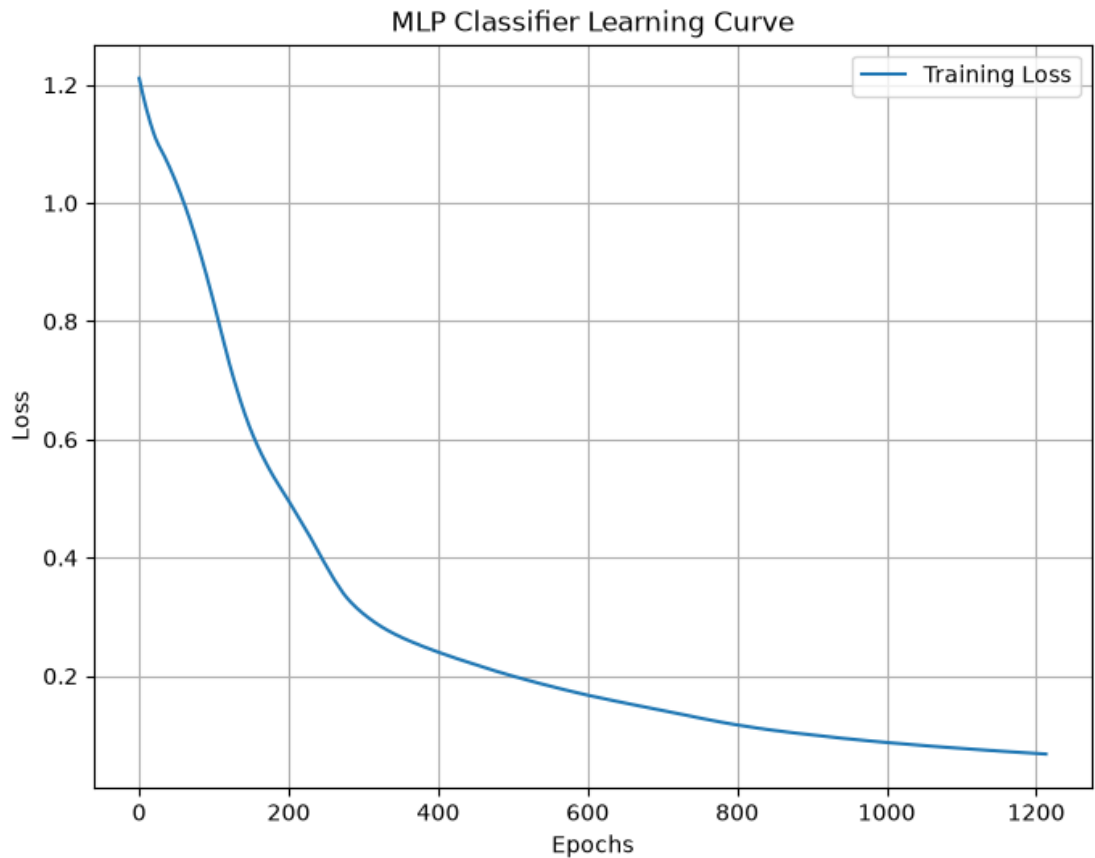
Number of layers: 6

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 1214



Umair Jamal (538263)

Hidden Layers: 2, Neurons: 20, Learning Rate: 0.001

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

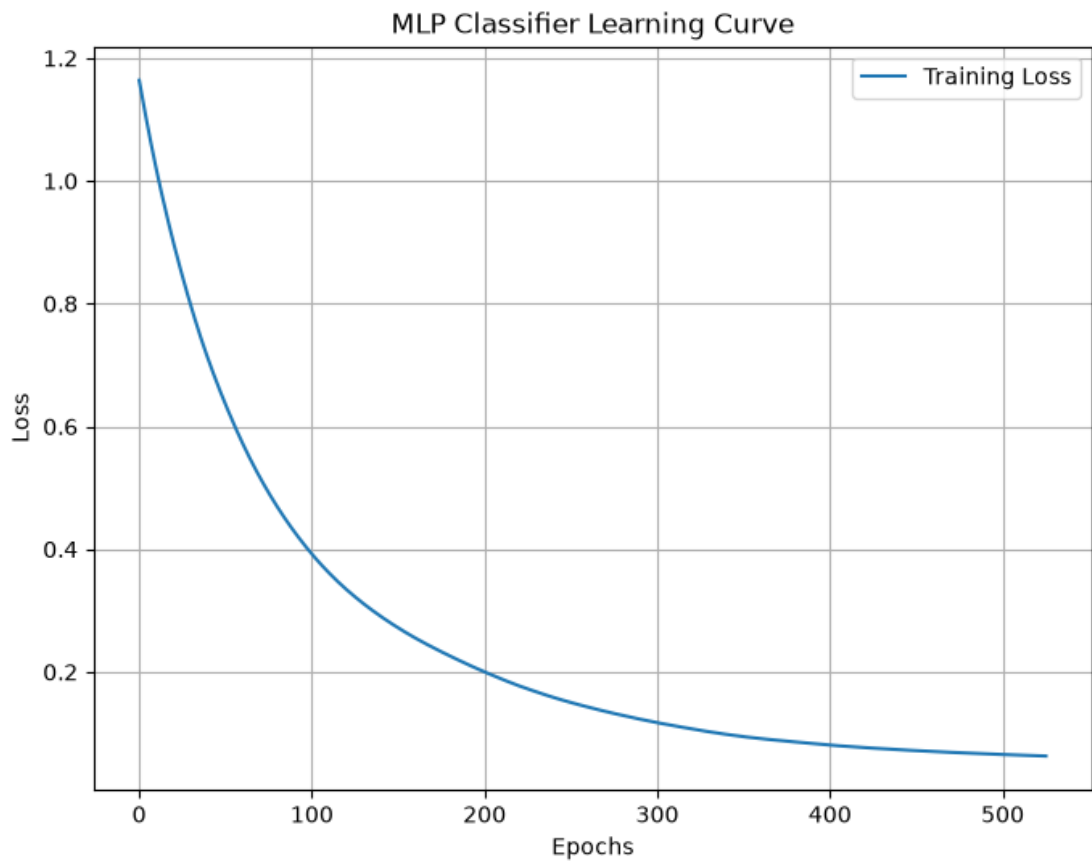
Number of layers: 4

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 526



Umair Jamal (538263)

Hidden Layers: 2, Neurons: 10, Learning Rate: 0.0005

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

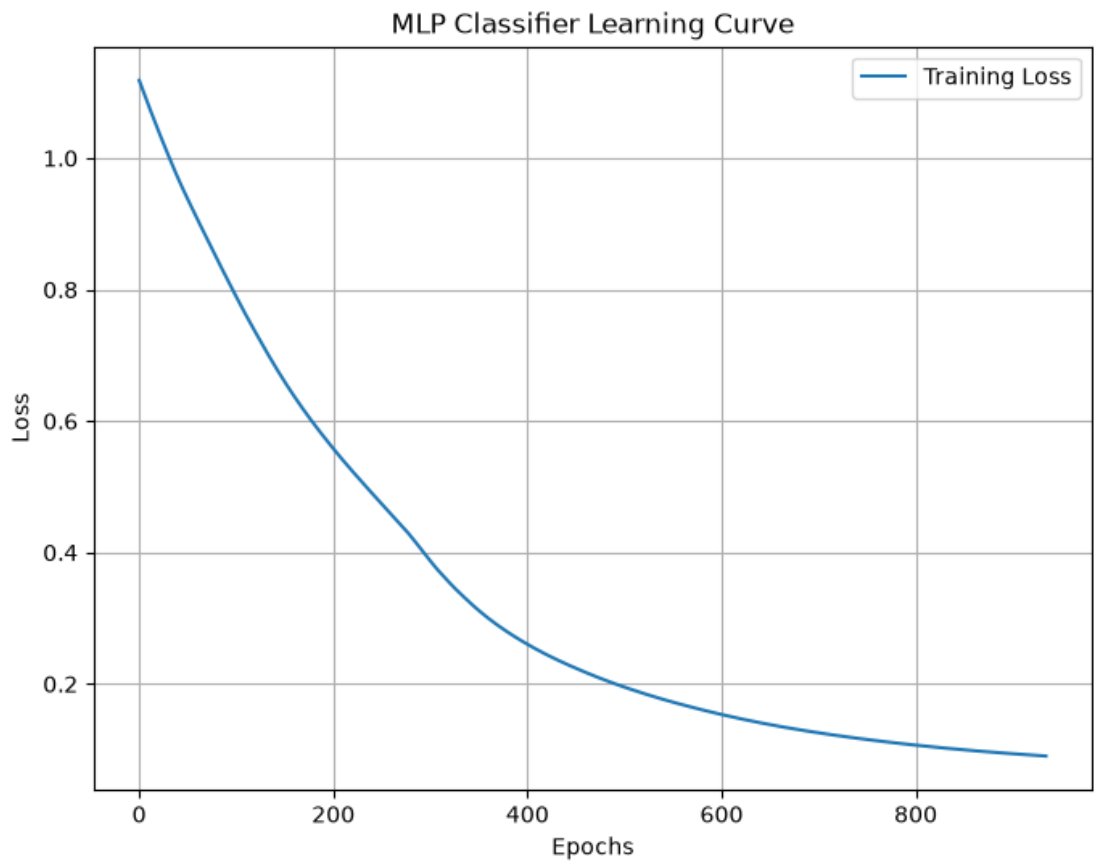
Number of layers: 4

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 935



Umair Jamal (538263)

Hidden Layers: 2, Neurons: 10, Learning Rate: 0.01

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

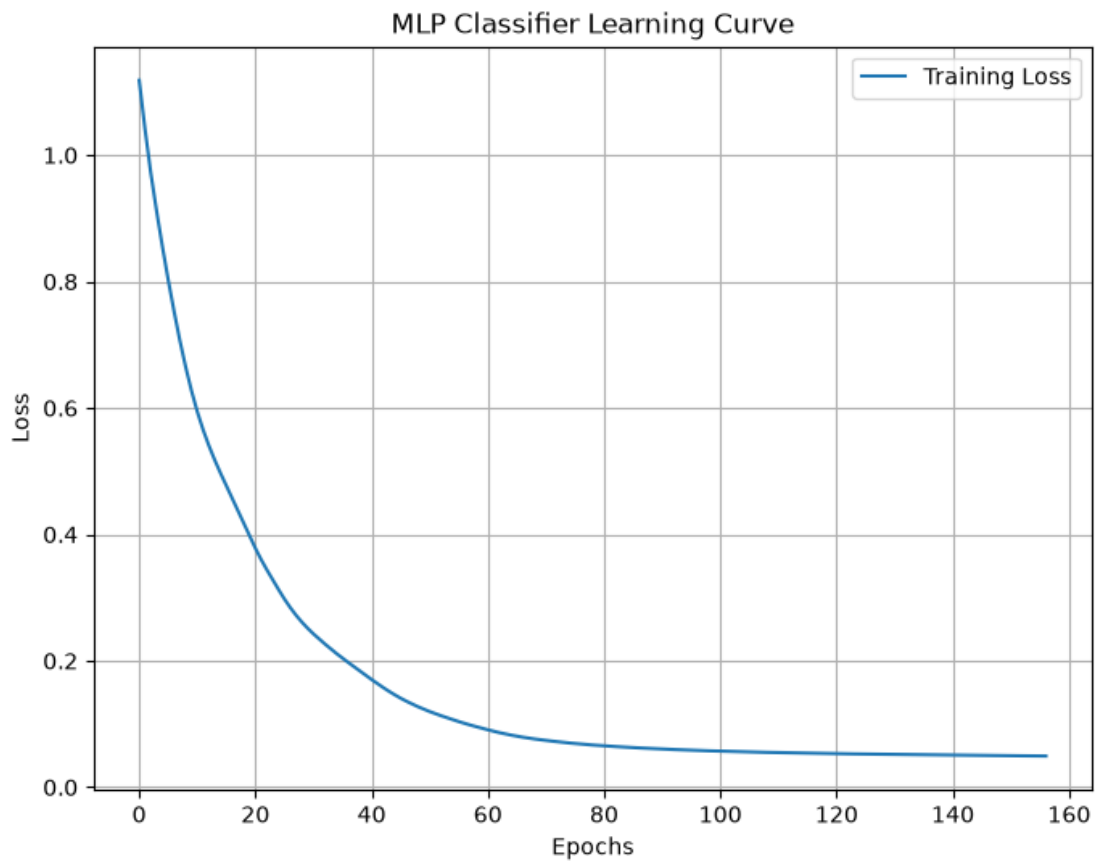
Number of layers: 4

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 157



Umair Jamal (538263)

Hidden Layers: 2, Neurons: 6, Learning Rate: 0.01

Accuracy: 1.00

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

MLP Structure:

Number of layers: 4

Number of outputs: 3

Activation function: relu

Output activation function: softmax

Number of epochs: 142

